MODEL LAB · 2026-10-14

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

27,888
held-out plate appearances

3.04%
held-out HR rate

0.5974
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-02, scored on the 27,888 that came after (actual HR rate 3.04%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5744	1.58×	0.02943	0.13530	0.00505
KNN comps Prior is the pooled rate of the nearest comparables.	0.5974	1.56×	0.02939	0.13431	0.00513
SVR Support-vector regression on the same feature matrix.	0.5901	1.50×	0.02944	0.13515	0.00667
Random forest Forest regression; a learned similarity metric.	0.5923	1.51×	0.02942	0.13464	0.00598
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5974	1.56×	0.02939	0.13431	0.00513
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5966	1.51×	0.02939	0.13431	0.00544
Linear regression Ordinary least squares on the same features.	0.5917	1.50×	0.02943	0.13478	0.00562
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5934	1.56×	0.02944	0.13481	0.00699
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5919	1.57×	0.02941	0.13464	0.00602
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5924	1.47×	0.02945	0.13487	0.00700

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.70** SVR **0.00** RF **0.30**

SHRINKAGE K PER PRIOR

league **255** knn **255** svr **255** rf **255** core **255** ensemble **255** linear **255** logistic **255** nn **255** xgb **255**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,279 PA, actual HR rate 2.29%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5776	1.33×	0.02238	0.10950	0.01207
KNN comps Prior is the pooled rate of the nearest comparables.	0.5726	1.73×	0.02237	0.10920	0.01260
SVR Support-vector regression on the same feature matrix.	0.5699	1.60×	0.02258	0.11313	0.01695
Random forest Forest regression; a learned similarity metric.	0.5932	1.47×	0.02254	0.10986	0.01824
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5726	1.73×	0.02237	0.10920	0.01260
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5808	1.60×	0.02240	0.10901	0.01445
Linear regression Ordinary least squares on the same features.	0.5880	1.47×	0.02258	0.11105	0.01856
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5926	1.60×	0.02261	0.11018	0.01912
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5864	1.73×	0.02242	0.10922	0.01573
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6051	1.87×	0.02273	0.11078	0.02337

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Teoscar Hernández 16.4%, Jackson Chourio 15.0%, Shohei Ohtani 14.4%, Mookie Betts 14.2%, Freddie Freeman 13.9%, William Contreras 13.2%
Flat league	6/6	Teoscar Hernández 15.6%, Jackson Chourio 14.9%, Mookie Betts 14.1%, Freddie Freeman 13.4%, Shohei Ohtani 13.1%, William Contreras 13.0%
KNN comps	6/6	Teoscar Hernández 16.5%, Jackson Chourio 15.1%, Shohei Ohtani 14.7%, Mookie Betts 14.0%, Freddie Freeman 13.8%, William Contreras 13.1%
SVR	6/6	Teoscar Hernández 16.4%, Jackson Chourio 15.5%, Mookie Betts 14.5%, Shohei Ohtani 14.5%, Freddie Freeman 13.4%, William Contreras 13.0%
Random forest	6/6	Teoscar Hernández 16.6%, Jackson Chourio 15.7%, Shohei Ohtani 14.8%, Mookie Betts 14.0%, Freddie Freeman 13.4%, William Contreras 13.1%
Core (KNN+SVR)	6/6	Teoscar Hernández 16.5%, Jackson Chourio 15.1%, Shohei Ohtani 14.7%, Mookie Betts 14.0%, Freddie Freeman 13.8%, William Contreras 13.1%
Forest blend	6/6	Teoscar Hernández 16.6%, Jackson Chourio 15.3%, Shohei Ohtani 14.8%, Mookie Betts 14.0%, Freddie Freeman 13.7%, William Contreras 13.1%
Linear regression	5/6	Teoscar Hernández 16.6%, Jackson Chourio 15.4%, Shohei Ohtani 14.5%, Mookie Betts 14.3%, Freddie Freeman 13.7%, Will Smith 13.1%
Logistic regression	6/6	Teoscar Hernández 16.6%, Jackson Chourio 15.4%, Shohei Ohtani 14.9%, Mookie Betts 14.3%, Freddie Freeman 13.5%, William Contreras 13.1%
Neural net (torch)	6/6	Teoscar Hernández 16.6%, Jackson Chourio 15.5%, Shohei Ohtani 14.8%, Mookie Betts 14.2%, Freddie Freeman 13.5%, William Contreras 13.1%
XGBoost	6/6	Teoscar Hernández 16.6%, Jackson Chourio 15.8%, Shohei Ohtani 15.1%, Mookie Betts 13.9%, Freddie Freeman 13.4%, William Contreras 13.2%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Teoscar Hernández**	LAD	**16.4%**	15.6%	16.5%	16.4%	16.6%	16.5%	16.6%	16.6%	16.6%	
2	**Jackson Chourio**	MIL	**15.0%**	14.9%	15.1%	15.5%	15.7%	15.1%	15.3%	15.4%	15.4%	
3	**Shohei Ohtani**	LAD	**14.4%**	13.1%	14.7%	14.5%	14.8%	14.7%	14.8%	14.5%	14.9%	
4	**Mookie Betts**	LAD	**14.2%**	14.1%	14.0%	14.5%	14.0%	14.0%	14.0%	14.3%	14.3%	
5	**Freddie Freeman**	LAD	**13.9%**	13.4%	13.8%	13.4%	13.4%	13.8%	13.7%	13.7%	13.5%	
6	**William Contreras**	MIL	**13.2%**	13.0%	13.1%	13.0%	13.1%	13.1%	13.1%	13.0%	13.1%	
7	**Andy Pages**	LAD	**12.9%**	12.4%	12.7%	12.3%	12.5%	12.7%	12.6%	12.3%	12.2%	
8	**Will Smith**	LAD	**12.5%**	12.2%	12.4%	12.2%	12.6%	12.4%	12.5%	13.1%	12.8%	
9	**Kyle Tucker**	LAD	**12.0%**	12.1%	11.9%	11.8%	12.0%	11.9%	12.0%	11.8%	11.6%	
10	**Jake Bauers**	MIL	**11.3%**	10.6%	11.8%	11.6%	11.5%	11.8%	11.7%	11.4%	11.4%	
11	**Brice Turang**	MIL	**10.9%**	11.2%	10.7%	10.8%	10.6%	10.7%	10.7%	10.3%	10.2%	
12	**Garrett Mitchell**	MIL	**9.2%**	8.7%	8.9%	8.0%	8.4%	8.9%	8.8%	8.4%	8.4%	
13	**Andrew Vaughn**	MIL	**8.5%**	9.6%	8.4%	7.6%	7.2%	8.4%	8.1%	7.8%	7.9%	
14	**Tommy Edman**	LAD	**7.5%**	8.3%	7.3%	7.5%	7.0%	7.3%	7.2%	7.5%	7.5%	
15	**Miguel Rojas**	LAD	**7.4%**	9.1%	7.9%	7.8%	7.7%	7.9%	7.8%	7.9%	8.1%	
16	**Joey Ortiz**	MIL	**5.8%**	6.8%	5.8%	6.0%	6.0%	5.8%	5.9%	5.8%	5.8%	
17	**Cooper Pratt**	MIL	**5.5%**	7.2%	5.5%	4.5%	5.3%	5.5%	5.4%	4.8%	5.0%	
18	**Luis Lara**	MIL	**4.6%**	6.2%	4.9%	3.5%	4.2%	4.9%	4.7%	3.6%	4.3%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-14 · model 2026.10.01-stack · built 2026-10-15 04:29:08 · live data 2026-10-15 04:29:08

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